

National Income Accounting: Equations and Study Tricks

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Part (a): Value Added Method, Contribution to GDP_{MP} and GDP_{FC}

Given Data

- Value of Output: Rs. 150 crores
- Intermediate Consumption: Rs. 50 crores
- Indirect Tax: Rs. 10 crores
- Subsidy: 0

Equations

1. **Gross Value Added at Market Price (GVA_{MP}):**

$$GVA_{MP} = \text{Value of Output} - \text{Intermediate Consumption}$$

$$GVA_{MP} = 150 - 50$$

2. **Contribution to GDP_{MP} (same as GVA_{MP}):**

$$GDP_{MP} = GVA_{MP}$$

$$GDP_{MP} = 150 - 50$$

3. **Contribution to GDP_{FC} :**

$$GDP_{FC} = GDP_{MP} - \text{Indirect Tax} + \text{Subsidy}$$

$$GDP_{FC} = (150 - 50) - 10 + 0$$

Part (a): National Income (Expenditure and Income Methods)

Given Data

- Consumption Expenditure: 500
- Investment Expenditure: 400
- Government Expenditure: 200
- Net Exports: 50
- Wages and Salaries: 600
- Rent: 100
- Interest: 150
- Profits: 200
- NFIA: -50
- Depreciation: 50
- Net Indirect Tax: 50

Expenditure Method

1. GDP_{MP} :

$$\text{GDP}_{\text{MP}} = \text{Consumption} + \text{Investment} + \text{Government Expenditure} + \text{Net Exports}$$

$$\text{GDP}_{\text{MP}} = 500 + 400 + 200 + 50$$

2. GNP_{MP} :

$$\text{GNP}_{\text{MP}} = \text{GDP}_{\text{MP}} + \text{NFIA}$$

$$\text{GNP}_{\text{MP}} = (500 + 400 + 200 + 50) + (-50)$$

3. **National Income** (NNP_{FC}):

$$\text{NNP}_{\text{FC}} = \text{GNP}_{\text{MP}} - \text{Depreciation} - \text{Net Indirect Tax}$$

$$\text{NNP}_{\text{FC}} = [(500 + 400 + 200 + 50) + (-50)] - 50 - 50$$

Income Method

1. GDP_{MP} :

$$GDP_{MP} = \text{Wages and Salaries} + \text{Rent} + \text{Interest} + \text{Profits} + \text{Net Indirect Tax} + \text{Depreciation}$$

$$GDP_{MP} = 600 + 100 + 150 + 200 + 50 + 50$$

2. GNP_{MP} :

$$GNP_{MP} = GDP_{MP} + NFIA$$

$$GNP_{MP} = (600 + 100 + 150 + 200 + 50 + 50) + (-50)$$

3. National Income (NNP_{FC}):

$$NNP_{FC} = GNP_{MP} - \text{Depreciation} - \text{Net Indirect Tax}$$

$$NNP_{FC} = [(600 + 100 + 150 + 200 + 50 + 50) + (-50)] - 50 - 50$$

Part (a): GDP_{MP} , GNP_{MP} , National Income (Expenditure Method)

Given Data

- Private Consumption Expenditure: 2000
- Government Consumption Expenditure: 500
- Net Factor Income from Abroad: -300
- Investment: 800
- Net Exports: 700
- Depreciation: 400
- Net Indirect Tax: 300

Equations

1. GDP_{MP} :

$$GDP_{MP} = \text{Private Consumption} + \text{Government Consumption} + \text{Investment} + \text{Net Exports}$$

$$GDP_{MP} = 2000 + 500 + 800 + 700$$

2. GNP_{MP} :

$$GNP_{MP} = GDP_{MP} + NFIA$$

$$GNP_{MP} = (2000 + 500 + 800 + 700) + (-300)$$

3. **National Income (NNP_{FC}):**

$$\begin{aligned}\text{NNP}_{\text{FC}} &= \text{GNP}_{\text{MP}} - \text{Depreciation} - \text{Net Indirect Tax} \\ \text{NNP}_{\text{FC}} &= [(2000 + 500 + 800 + 700) + (-300)] - 400 - 300\end{aligned}$$

Part (b): NNP , NDP , GDP , GNP

Given Data

- National Income: 13000
- Depreciation: 300
- NFIA: -480
- Indirect Taxes: 350

Equations

1. NNP_{FC} (National Income, given):

$$\text{NNP}_{\text{FC}} = 13000$$

2. NDP_{FC} :

$$\begin{aligned}\text{NDP}_{\text{FC}} &= \text{NNP}_{\text{FC}} - \text{NFIA} \\ \text{NDP}_{\text{FC}} &= 13000 - (-480)\end{aligned}$$

3. GDP_{MP} :

$$\begin{aligned}\text{GDP}_{\text{MP}} &= \text{NDP}_{\text{FC}} + \text{Depreciation} + \text{Indirect Taxes} \\ \text{GDP}_{\text{MP}} &= [13000 - (-480)] + 300 + 350\end{aligned}$$

4. GNP_{MP} :

$$\begin{aligned}\text{GNP}_{\text{MP}} &= \text{GDP}_{\text{MP}} + \text{NFIA} \\ \text{GNP}_{\text{MP}} &= \{[13000 - (-480)] + 300 + 350\} + (-480)\end{aligned}$$

Part (b): NDP (Income and Expenditure Methods)

Given Data

- Consumption: 3000
- Investment Expenditure: 2000
- Government Expenditure: 700

- Exports: 600
- Imports: 300
- Intermediate Consumption: 2000
- Wages and Salaries: 2000
- Rent: 500
- Interest: 500
- Profit: 1000

Expenditure Method

1. GDP_{MP} :

$$GDP_{MP} = \text{Consumption} + \text{Investment} + \text{Government Expenditure} + (\text{Exports} - \text{Imports})$$

$$GDP_{MP} = 3000 + 2000 + 700 + (600 - 300)$$

2. NDP_{FC} :

$$NDP_{FC} = GDP_{MP} - \text{Intermediate Consumption}$$

$$NDP_{FC} = [3000 + 2000 + 700 + (600 - 300)] - 2000$$

Income Method

1. NDP_{FC} :

$$NDP_{FC} = \text{Wages and Salaries} + \text{Rent} + \text{Interest} + \text{Profit}$$

$$NDP_{FC} = 2000 + 500 + 500 + 1000$$

Part (b): GNP, NNP, National Income

Given Data

- GDP: 5000 (in 100 billion)
- NFIA: -50
- Indirect Tax: 70
- Subsidies: 20
- Depreciation: 30

Equations

1. GNP_{MP} :

$$\text{GNP}_{\text{MP}} = \text{GDP}_{\text{MP}} + \text{NFIA}$$

$$\text{GNP}_{\text{MP}} = 5000 + (-50)$$

2. NNP_{MP} :

$$\text{NNP}_{\text{MP}} = \text{GNP}_{\text{MP}} - \text{Depreciation}$$

$$\text{NNP}_{\text{MP}} = [5000 + (-50)] - 30$$

3. National Income (NNP_{FC}):

$$\text{NNP}_{\text{FC}} = \text{NNP}_{\text{MP}} - (\text{Indirect Tax} - \text{Subsidies})$$

$$\text{NNP}_{\text{FC}} = \{[5000 + (-50)] - 30\} - (70 - 20)$$

Part (b): National Income (Income Method), NDP, NNP

Given Data

- Wages and Salaries: 800
- Rent: 300
- Depreciation: 200
- Interest: 400
- Net Indirect Tax: 400
- NFIA: 100
- Profit: 400

Equations

1. National Income (NNP_{FC}):

$$\text{NNP}_{\text{FC}} = \text{Wages and Salaries} + \text{Rent} + \text{Interest} + \text{Profit}$$

$$\text{NNP}_{\text{FC}} = 800 + 300 + 400 + 400$$

2. NDP_{FC} :

$$\text{NDP}_{\text{FC}} = \text{NNP}_{\text{FC}} - \text{NFIA}$$

$$\text{NDP}_{\text{FC}} = (800 + 300 + 400 + 400) - 100$$

3. NNP_{MP} :

$$NNP_{MP} = NNP_{FC} + \text{Net Indirect Tax}$$
$$NNP_{MP} = [(800 + 300 + 400 + 400) - 100] + 400$$

Easy Trick to Study and Solve Similar Problems

Trick 1: Memorize Key Formulas with Acronyms

- **Expenditure Method (CIGNET):**

$$GDP_{MP} = C \text{ (Consumption)} + I \text{ (Investment)} + G \text{ (Government Expenditure)} + NET \text{ (Net Exports)}$$

- **Income Method (WRIP):**

$$NDP_{FC} = W \text{ (Wages and Salaries)} + R \text{ (Rent)} + I \text{ (Interest)} + P \text{ (Profit)}$$

- **Adjustments:**

$$GNP_{MP} = GDP_{MP} + NFIA$$
$$NNP_{MP} = GNP_{MP} - \text{Depreciation}$$
$$NNP_{FC} = NNP_{MP} - (\text{Indirect Tax} - \text{Subsidies})$$

Trick 2: Use a Step-by-Step Checklist

1. Identify the Method:

- Expenditure: Look for C, I, G, Exports, Imports.
- Income: Look for Wages, Rent, Interest, Profits.
- Value Added: Look for Output and Intermediate Consumption.

2. Calculate GDP_{MP} or NDP_{FC} First:

$$GDP_{MP} = C + I + G + (X - M) \quad (\text{Expenditure})$$
$$NDP_{FC} = \text{Wages} + \text{Rent} + \text{Interest} + \text{Profits} \quad (\text{Income})$$
$$GVA_{MP} = \text{Output} - \text{Intermediate Consumption} \quad (\text{Value Added})$$

3. Adjust for NFIA:

$$GNP_{MP} = GDP_{MP} + NFIA$$

4. Adjust for Depreciation:

$$NNP_{MP} = GNP_{MP} - \text{Depreciation}$$

5. Adjust for Net Indirect Tax:

$$NNP_{FC} = NNP_{MP} - (\text{Indirect Tax} - \text{Subsidies})$$
$$GDP_{FC} = GDP_{MP} - (\text{Indirect Tax} - \text{Subsidies})$$

Trick 3: Practice with a Flowchart

- Start with GDP_{MP} or NDP_{FC} .
- Add NFIA $\rightarrow GNP_{MP}$.
- Subtract Depreciation $\rightarrow NNP_{MP}$.
- Subtract Net Indirect Tax $\rightarrow NNP_{FC}$ (National Income).

Trick 4: Use a Table for Data Organization

Component	Value
Consumption (C)	
Investment (I)	
Government Exp (G)	
Net Exports (X - M)	
NFIA	
Depreciation	
Net Indirect Tax	

Trick 5: Memorize Key Relationships

- GDP_{MP} to NNP_{FC} :

$$NNP_{FC} = GDP_{MP} + NFIA - \text{Depreciation} - \text{Net Indirect Tax}$$

- NDP_{FC} to GDP_{MP} :

$$GDP_{MP} = NDP_{FC} + \text{Depreciation} + \text{Net Indirect Tax}$$

Trick 6: Practice with Hypothetical Numbers

Assign simple numbers (e.g., $C = 100$, $I = 50$, $G = 30$, Net Exports = 20) and calculate GDP_{MP} , GNP_{MP} , and NNP_{FC} .